DRUGS · ALLERGY, RHINITIS AND BIOLOGICS

# Oral antihistamines

Oral H1 antihistamines relieve the histamine-driven symptoms of allergic rhinitis (sneezing, itch, watery rhinorrhoea and eye symptoms) within an hour, but do little for congestion. The second-generation drugs (cetirizine, loratadine, fexofenadine and their metabolites) are non-sedating or minimally sedating and are the ones to prescribe. The first-generation drugs such as chlorphenamine cross the blood-brain barrier, sedate and have anticholinergic effects.

## 1 CHAPTER ONE At a glance

The drugs on one page.

### 1.1 Summary

ORAL ANTIHISTAMINES

| | SUMMARY |
|---|---|
| **Agents** | Second generation: cetirizine, levocetirizine, loratadine, desloratadine, fexofenadine, bilastine. First generation: chlorphenamine, promethazine, hydroxyzine |
| **How they work** | Inverse agonists at the H1 receptor; stabilise it in its inactive state |
| **ENT uses** | Mild or intermittent allergic rhinitis; add-on when a nasal steroid is not enough; allergic conjunctivitis; urticaria and angioedema (histaminergic); itch |
| **Adult dose** | Cetirizine 10 mg once daily; loratadine 10 mg once daily; fexofenadine 120 mg once daily (180 mg for urticaria); chlorphenamine 4 mg every 4 to 6 hours (maximum 24 mg a day) |
| **Child dose** | Cetirizine 2 to 5 years 2.5 mg twice daily, 6 to 11 years 5 mg twice daily; loratadine 2 to 11 years 5 mg once daily (under 31 kg), 10 mg if heavier; fexofenadine 6 to 11 years 30 mg twice daily |
| **Avoid** | First-generation agents in the elderly, drivers and machine operators, glaucoma, prostatic hypertrophy and epilepsy |

| | SUMMARY |
|---|---|
| **Pregnancy** | Loratadine and cetirizine have the most data and are preferred if needed |

## 2 CHAPTER TWO How they work

Mechanism and handling.

### 2.1 Generations

Second-generation drugs are substrates of P-glycoprotein and are excluded from the brain, so they cause little sedation (cetirizine sedates a minority; fexofenadine is least sedating). First-generation drugs cross freely, sedate, impair driving even when the patient feels alert, and block muscarinic, alpha-adrenergic and serotonin receptors.

## 3 CHAPTER THREE Using them in ENT

Indications and doses.

### 3.1 Allergic rhinitis

ARIA guidance: an oral antihistamine is reasonable for mild intermittent symptoms; for persistent or moderate to severe disease, a nasal steroid is more effective, and an intranasal antihistamine-steroid combination comes before adding an oral antihistamine. Oral antihistamines are useful for prominent eye symptoms and for patients who cannot use sprays.

### 3.2 Angioedema and anaphylaxis

In anaphylaxis, **adrenaline** is the treatment; antihistamines are third-line and only treat skin symptoms. In histaminergic angioedema without airway involvement, a non-sedating antihistamine at up to four times the licensed dose (specialist advice)

is used. Antihistamines do not work in **bradykinin-mediated** angioedema (ACE inhibitor or hereditary).

# 4 CHAPTER FOUR Safety

What goes wrong.

## 4.1 Adverse effects

- Second generation: headache, dry mouth, occasional drowsiness (cetirizine).
- First generation: **sedation**, psychomotor impairment, anticholinergic effects (urinary retention, confusion and falls in the elderly), paradoxical excitation in children.
- Older drugs such as terfenadine and astemizole caused QT prolongation and were withdrawn.

## 4.2 Interactions

Alcohol and other sedatives (first generation); fruit juices reduce fexofenadine absorption; antacids with aluminium and magnesium reduce fexofenadine absorption.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- First- versus second-generation antihistamines.
- Which symptoms antihistamines do and do not treat.

- Their limited place in **anaphylaxis** and none in bradykinin angioedema.

## 5.2 Common pitfalls

COMMON PITFALLS

- Prescribing chlorphenamine for daily allergic rhinitis.
- Relying on antihistamines in anaphylaxis.
- Treating ACE inhibitor angioedema with antihistamines.

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